



Design of Circular Corrugated Horn with Constant Width Corrugations

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DESIGN OF CIRCULAR CORRUGATED HORN WITH CONSTANT WIDTH CORRUGATIONS

Outline

- ❖ Introduction
- ❖ Platelet horns
- ❖ Design
- ❖ Fabrication
- ❖ Test
- ❖ Conclusions



DESIGN OF CIRCULAR CORRUGATED HORN WITH CONSTANT WIDTH CORRUGATIONS

INTRODUCTION

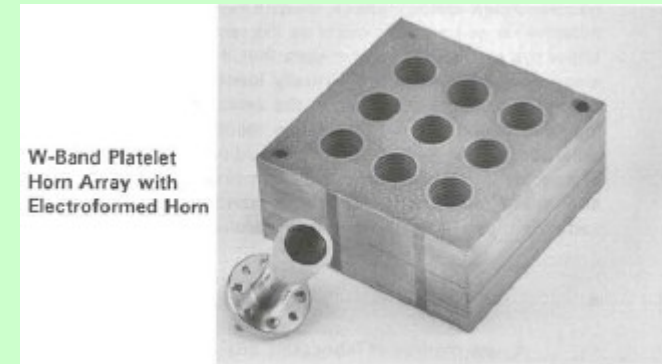
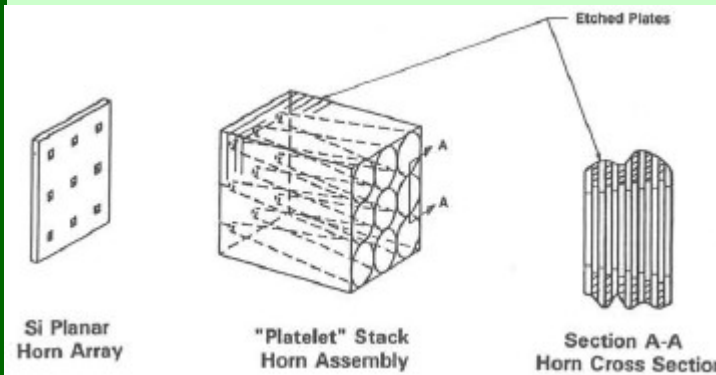
- ❖ Modern focal plane arrays for reflector antennas require:
 - ❑ Large number of elements
 - ❑ High performances
 - ❑ Reduced dimension and weight
 - ❑ Reliable fabrication technique
 - ❑ Low cost

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PLATELET HORNS

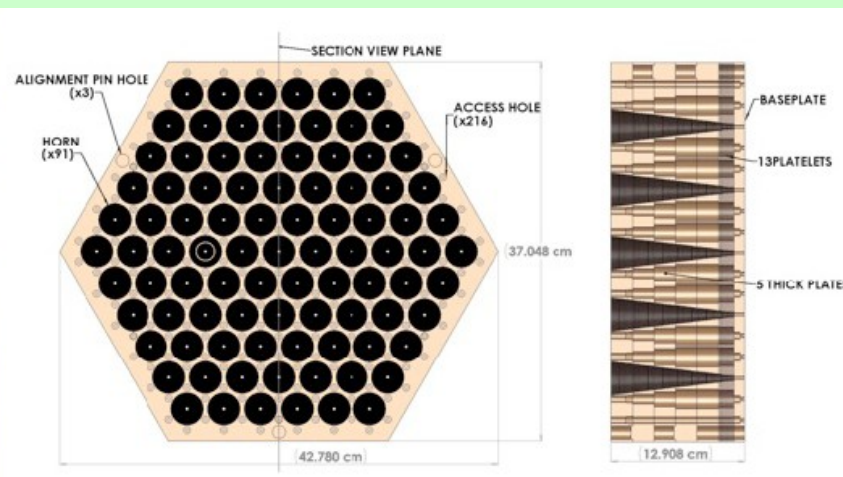
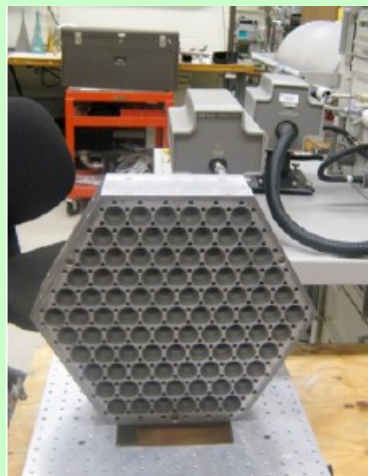
Stacks of photoetched *platelets*

R. W. Haas, D. Brest, H. Mueggenburg, L. Lang, D. Heimlich, "Fabrication and Performance of MMW and SMMW Platelet Horn Arrays," *International Journal of Infrared and Millimeter Waves*, vol. 14, pp. 2289-2293, 1993



Stacks of machined thick and thin *platelets*

J. Gundersen, E. Wollack, "Millimeter Wave Corrugated Platelet Feeds," 2009 J. Phys.: Conf. Ser. 155, 012005.



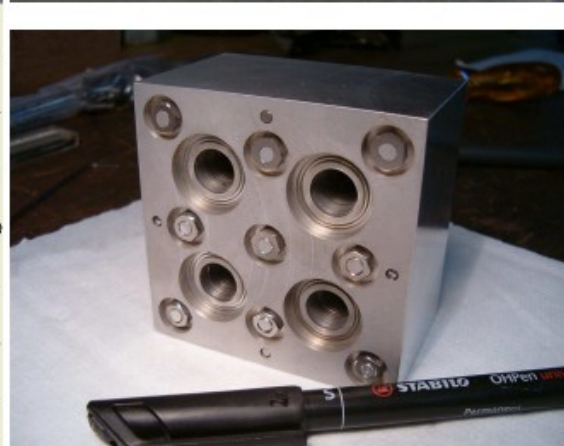
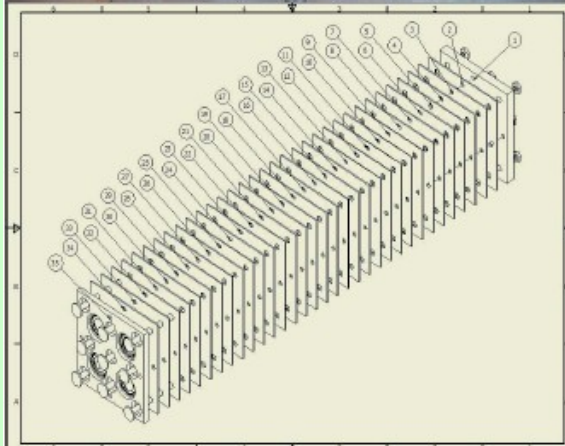
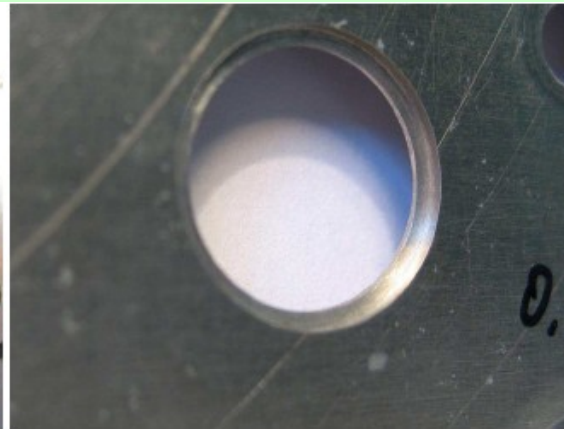
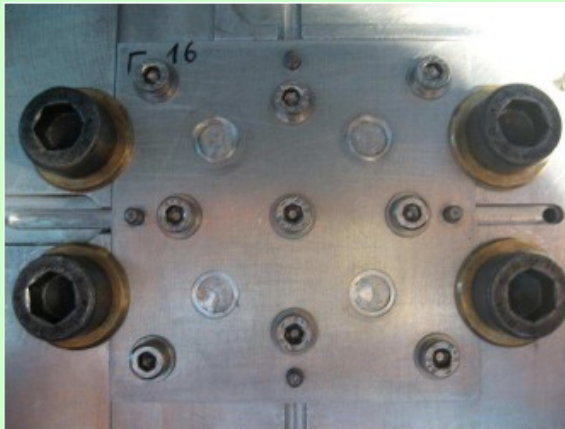


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PLATELET HORNS

Stacks of machined non bonded *platelets*

F. Del Torto et al. "W-band prototype of platelet feed-horn array for CMB polarisation measurements", JINST, Vol. 6, June 2011



.... and few
other works



DESIGN OF CIRCULAR CORRUGATED HORN WITH CONSTANT WIDTH CORRUGATIONS

DESIGN

- ❖ Requirements (performances):
 - ❑ Bandwidth: 33-50GHz (Q-band)
 - ❑ Return loss: >30dB
 - ❑ Cross-polarization: <-30dB
 - ❑ Tapering: 11dB@12°

- ❖ Requirements (mechanics):
 - ❑ High compactness (array oriented)
 - ❑ Constant step width: ($w = 2.6\text{mm}$)

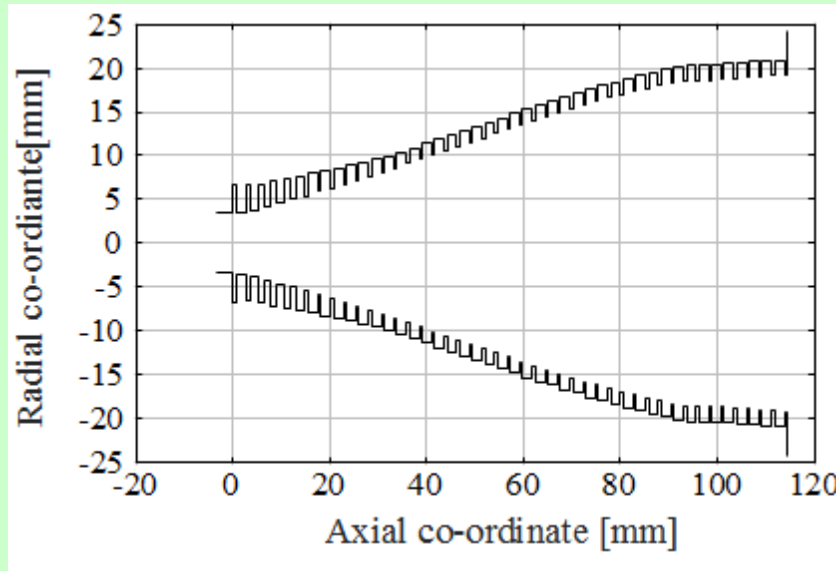


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DESIGN

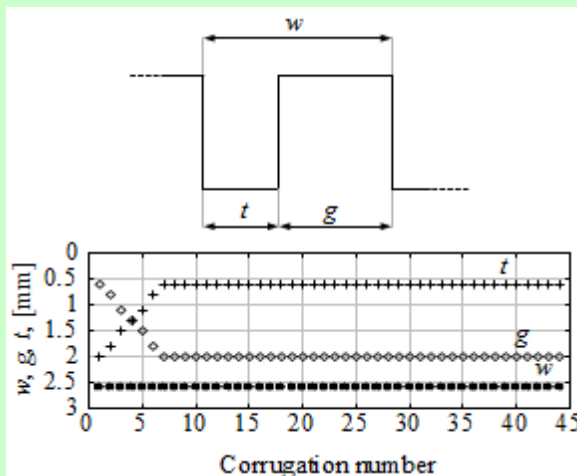
- ❖ CAD procedure:
 - ❑ Analysis
 - ❑ Proprietary software
 - ❑ Mode Matching
 - ❑ Method of Moments
 - ❑ Optimization
 - ❑ Quasi Newton Method

DESIGN



Corrugation profile

Optimized sine squared shape



Corrugation geometry

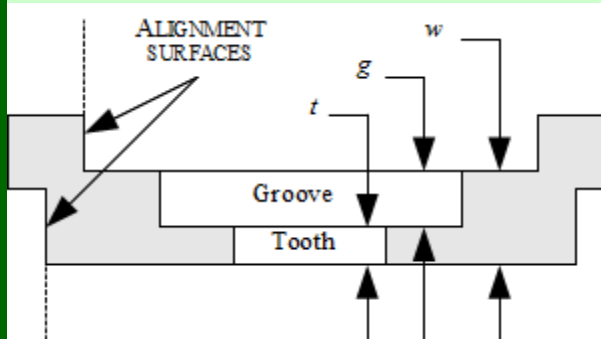
Constant step width ($w = g + t$): *To be noticed*

A total of $N_T = 44$ corrugations

Ideally $N_T = 44$ equally thick plates

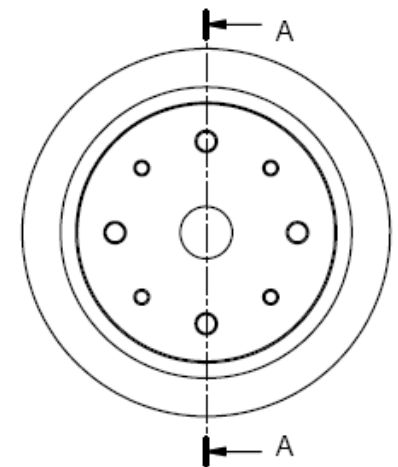
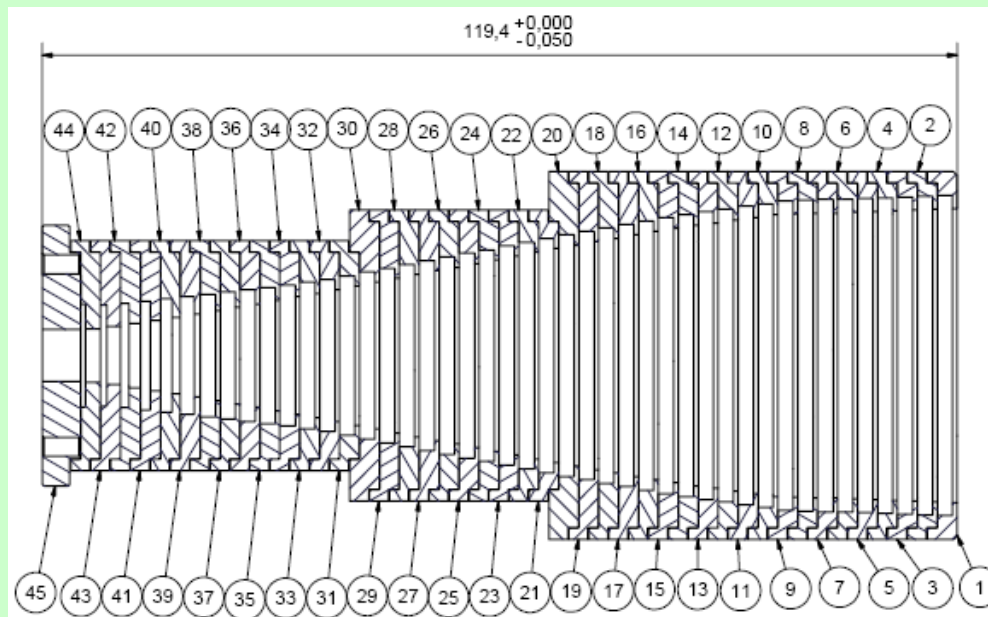
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FABRICATION



Self aligned disks for each horn corrugation
 Instead of platelets with whole array pattern of holes
 No pins; No glue; No bonding;
 Only tightening screws;
 Allowing consistent mass reduction
 Allowing array modularity

Single horn

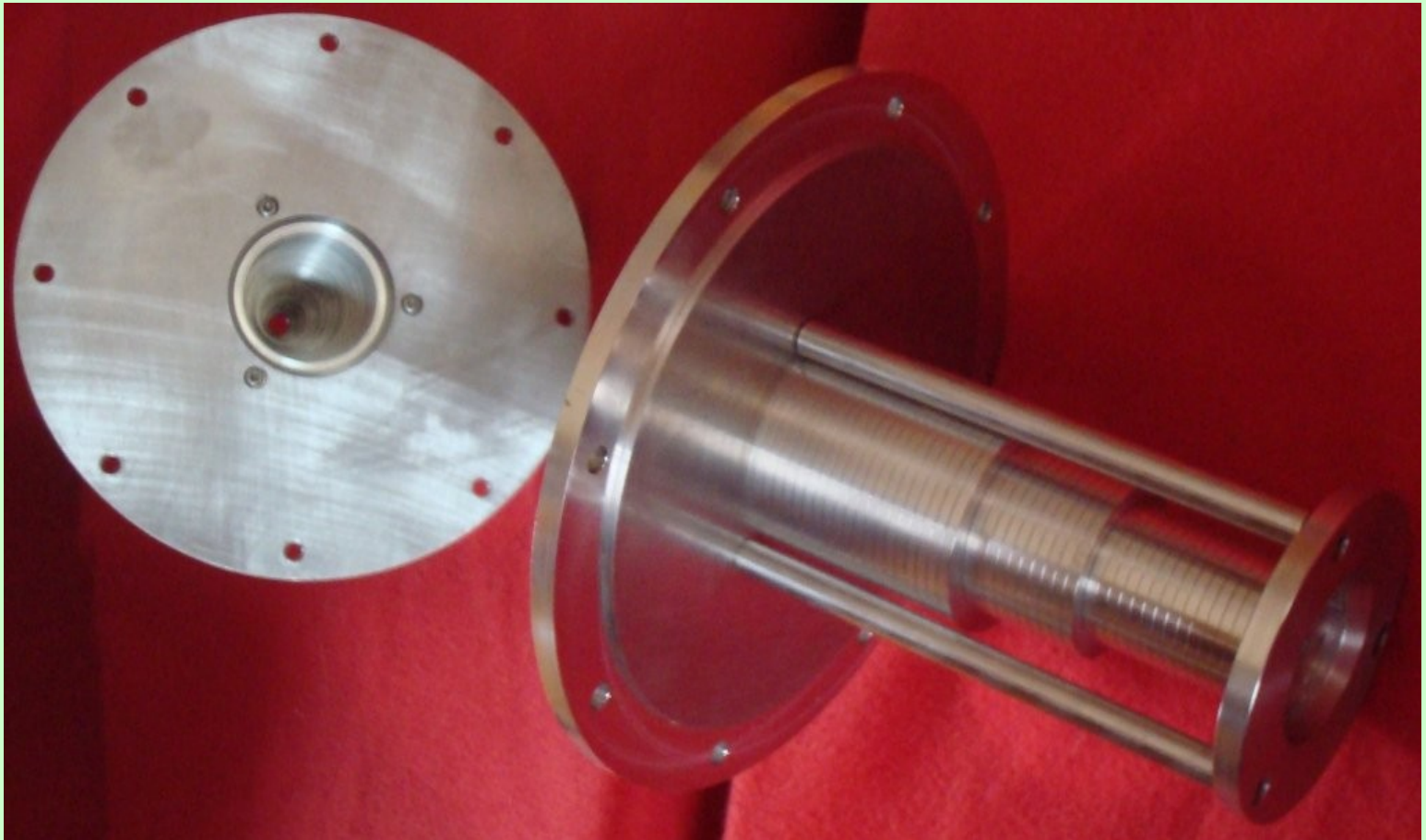




DESIGN OF CIRCULAR CORRUGATED HORN WITH CONSTANT WIDTH CORRUGATIONS

FABRICATION

Single horn (mirrored)



FABRICATION

Array of seven horns

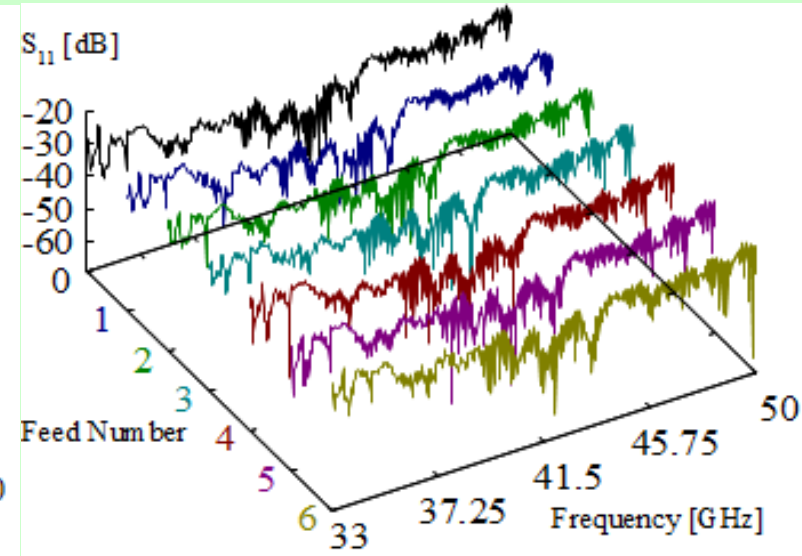
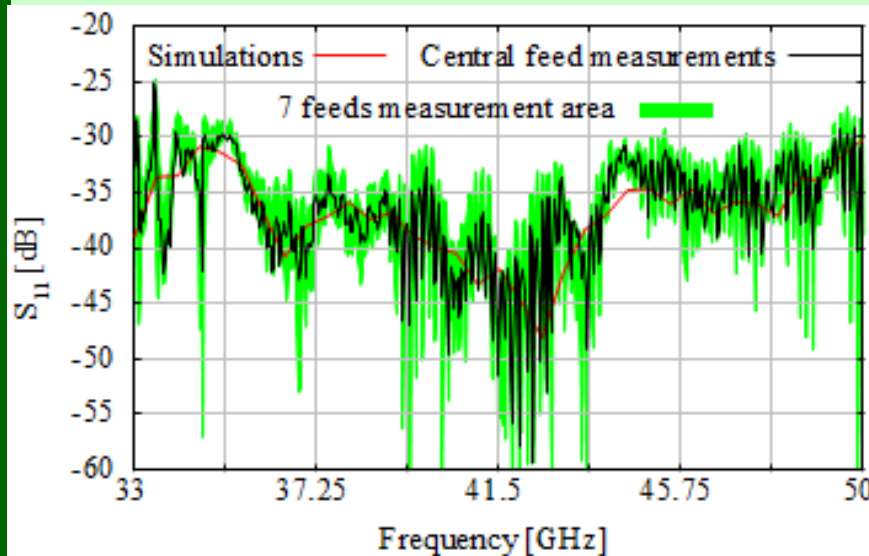




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TEST

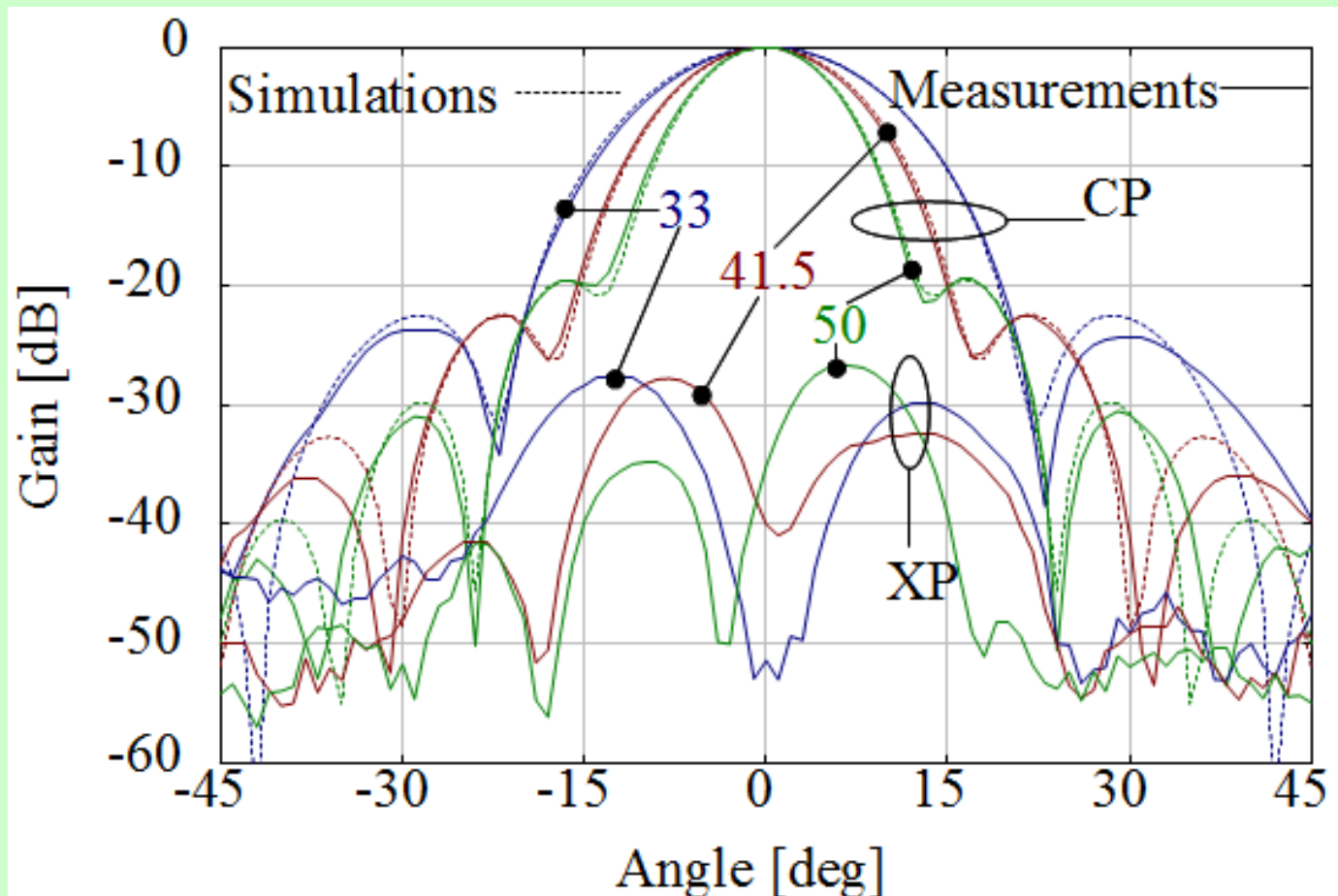
Reflection coefficient



Cross-Coupling $< -60dB$ (instrument limit)

TEST

Co-polar (comparison) and cross-polar pattern in the 45° cut at three frequencies

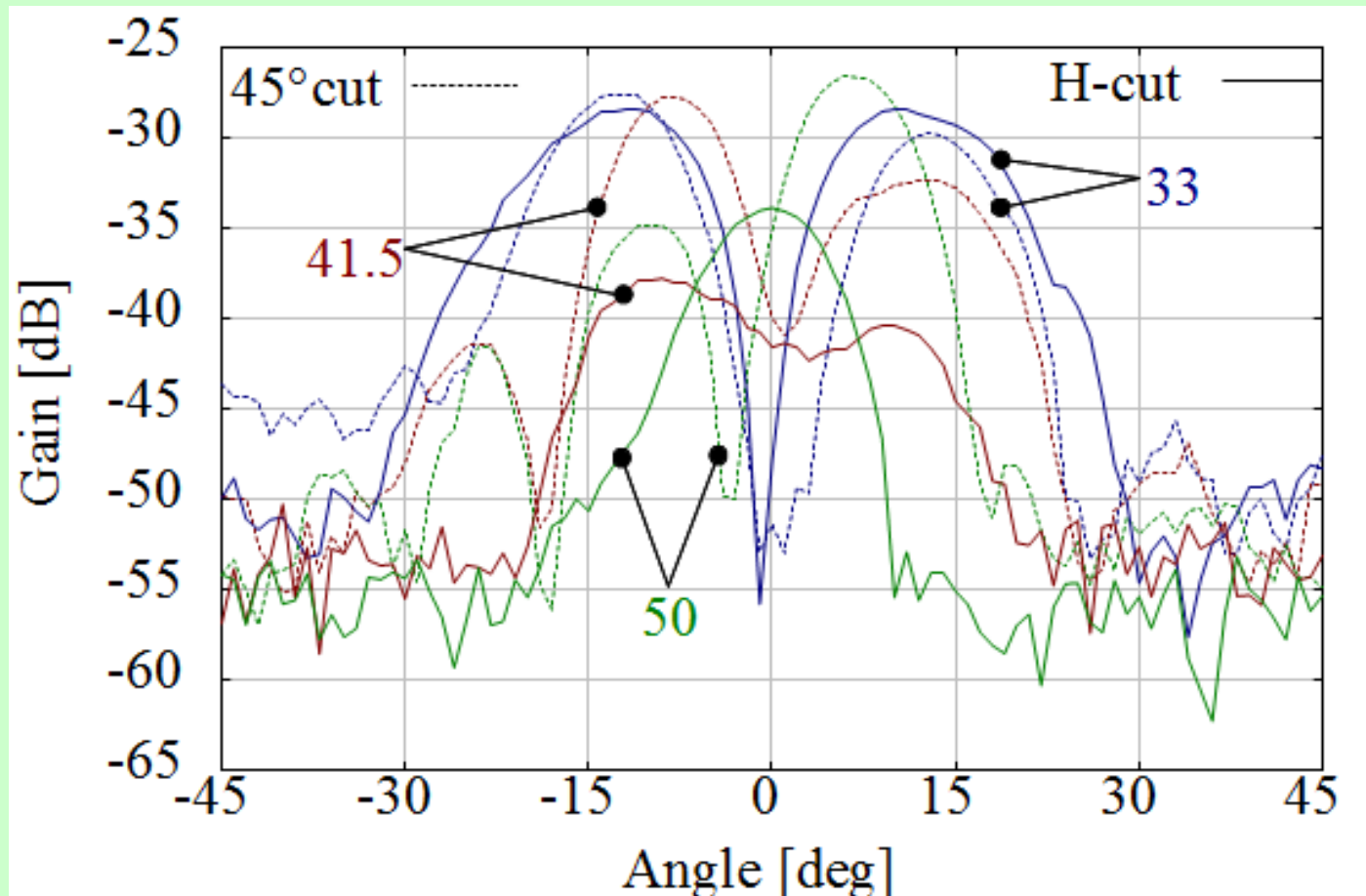




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TEST

Measured cross-polar pattern in the 45° and H cuts at three frequencies

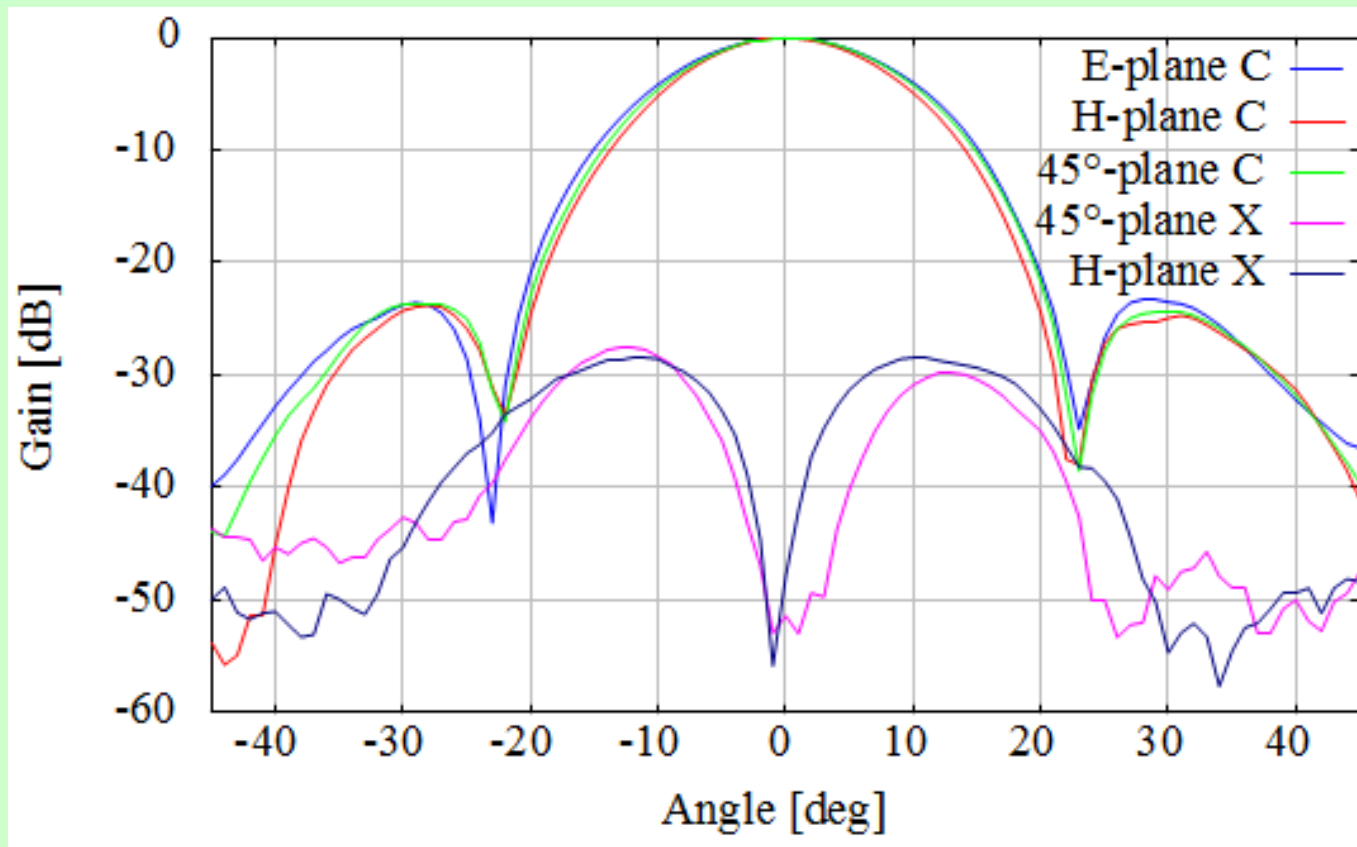




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TEST

Measured co-polar and cross-polar pattern at 33GHz

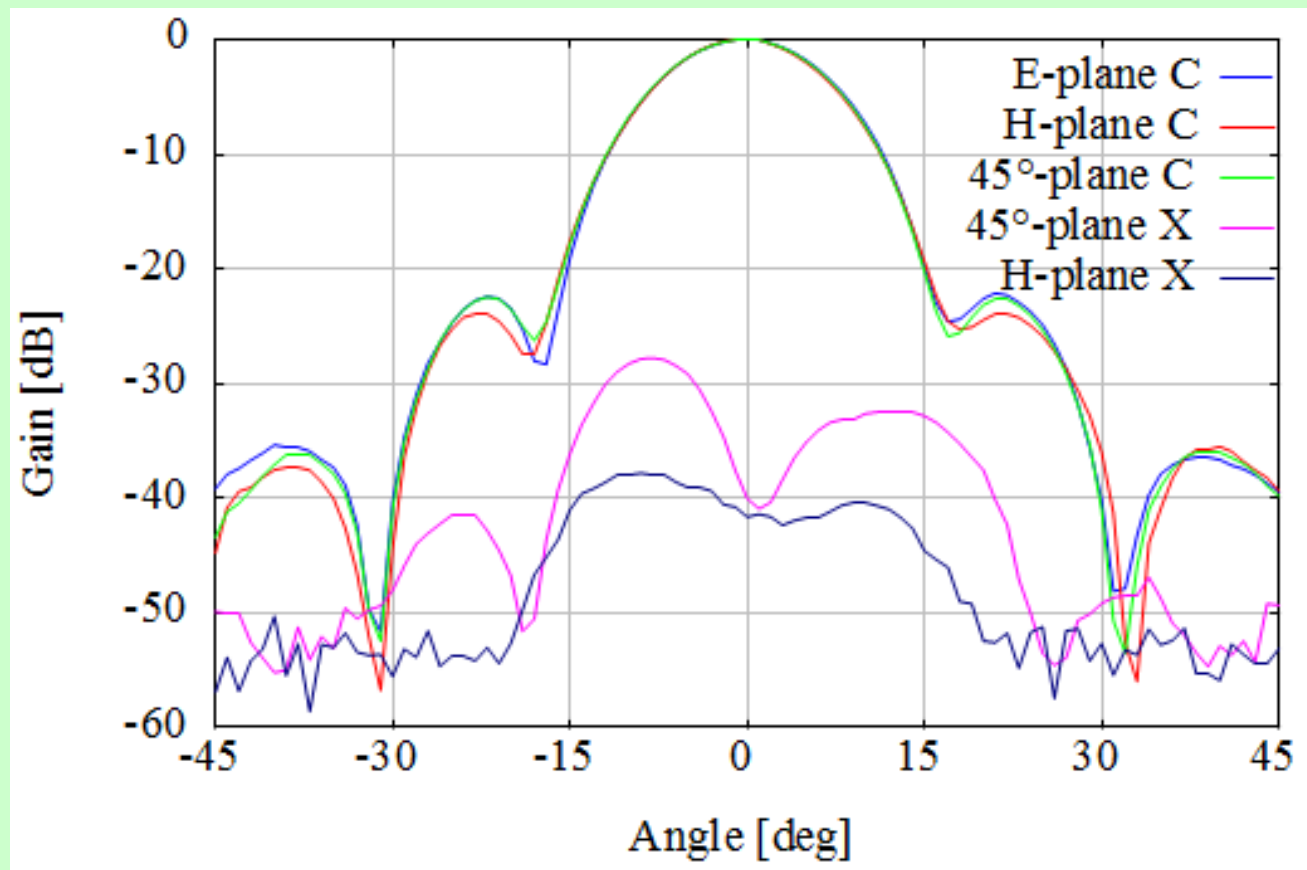




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TEST

Measured co-polar and cross-polar pattern at 41.5GHz

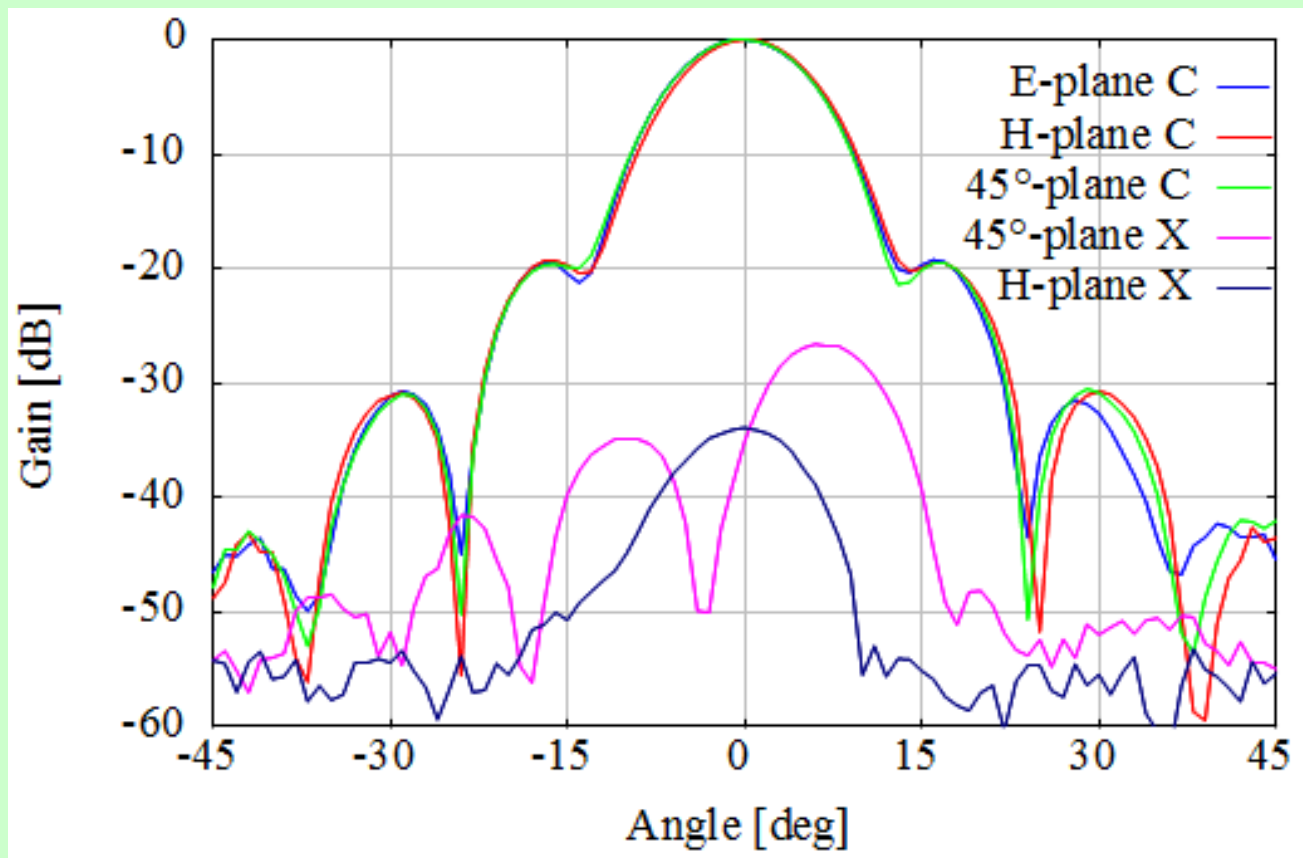




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TEST

Measured co-polar and cross-polar pattern at 50GHz





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CONCLUSIONS

- ❖ A corrugated horn design with constant corrugation width for platelet technology and low-cost production has been presented
- ❖ The electromagnetic tests highlight very good performances at Q band
- ❖ Proposed design and technology show attractive features for high-performance and low-cost focal plane arrays in the lower part of the millimetre-wave band